

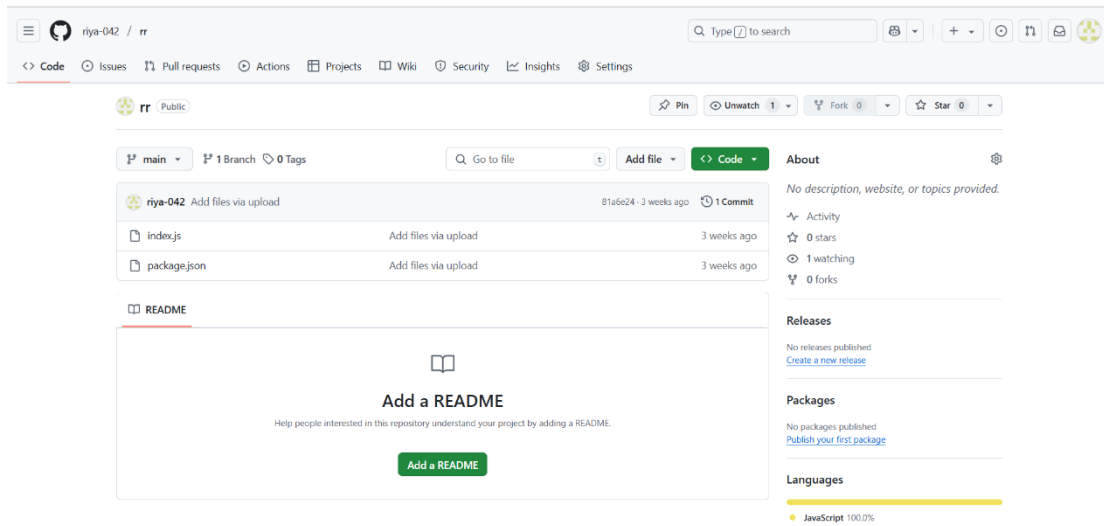
Assignment No: 10

Deploy a project from GitHub to EC2 by creating a new security group and user data.

Solution:

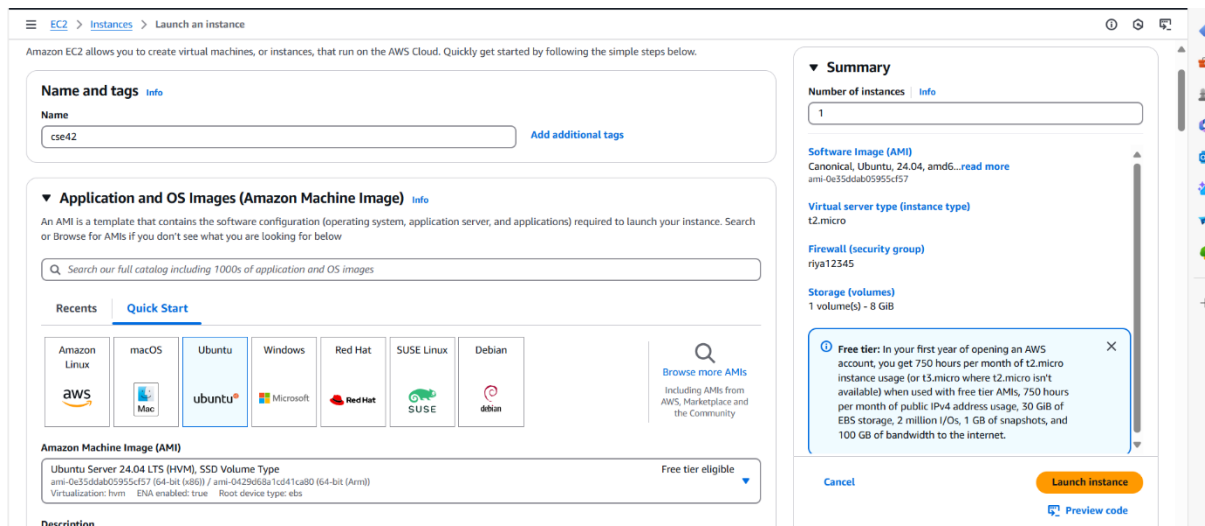
Step 1:

Upload required files to GitHub

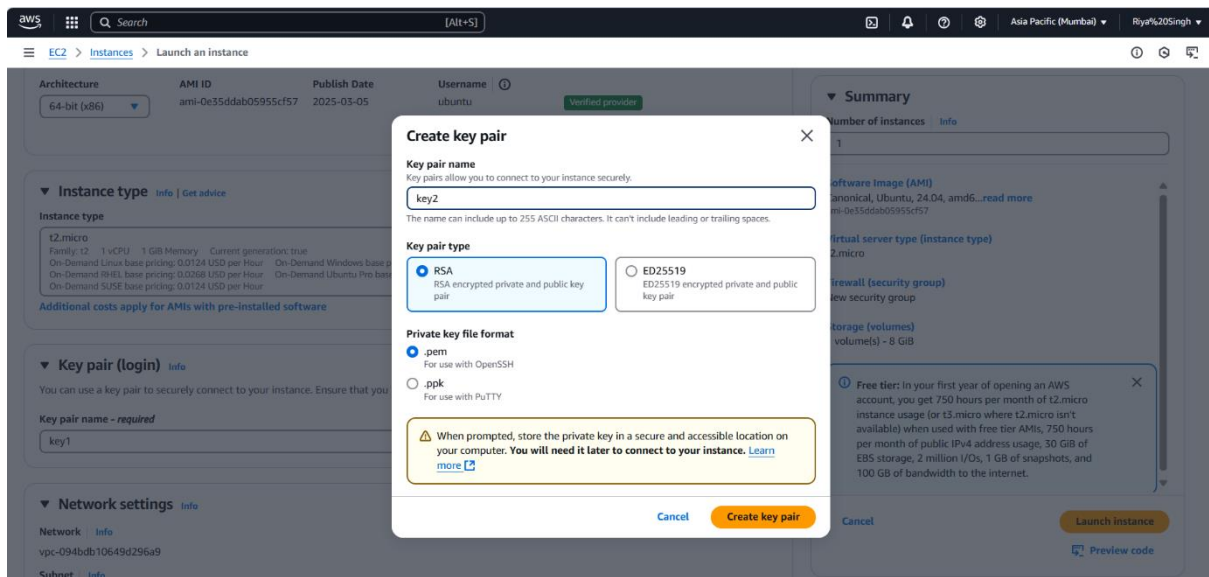


Step 2:

Log into AWS and open EC2. Click on launch instance. Give a name to the instance and select the operating system as Ubuntu.



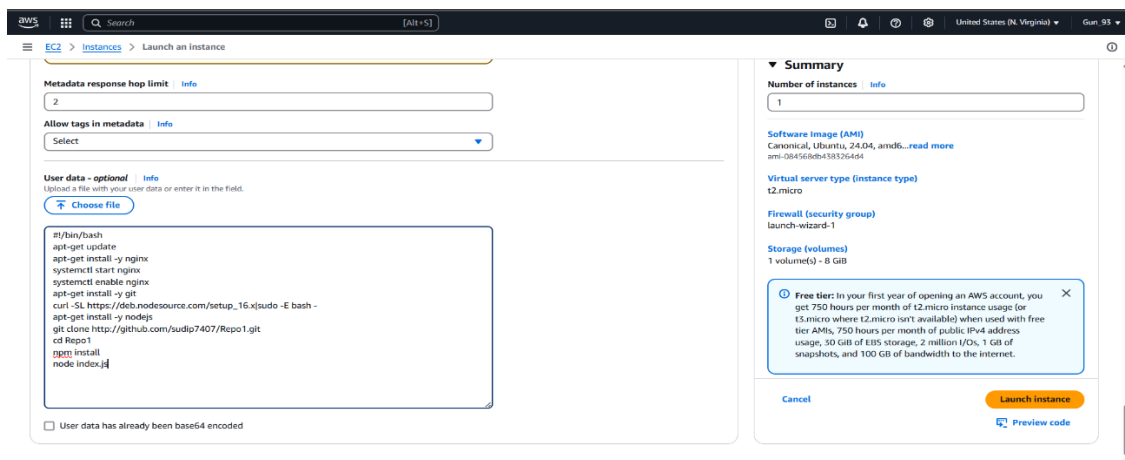
Step 3: Create Key Pair.



Step 4:

From the Network Section Select Existing Security Group then choose your previously created security group.

Step 5: After Selecting security group go to the user data section and write given cmd. and click launch instance.



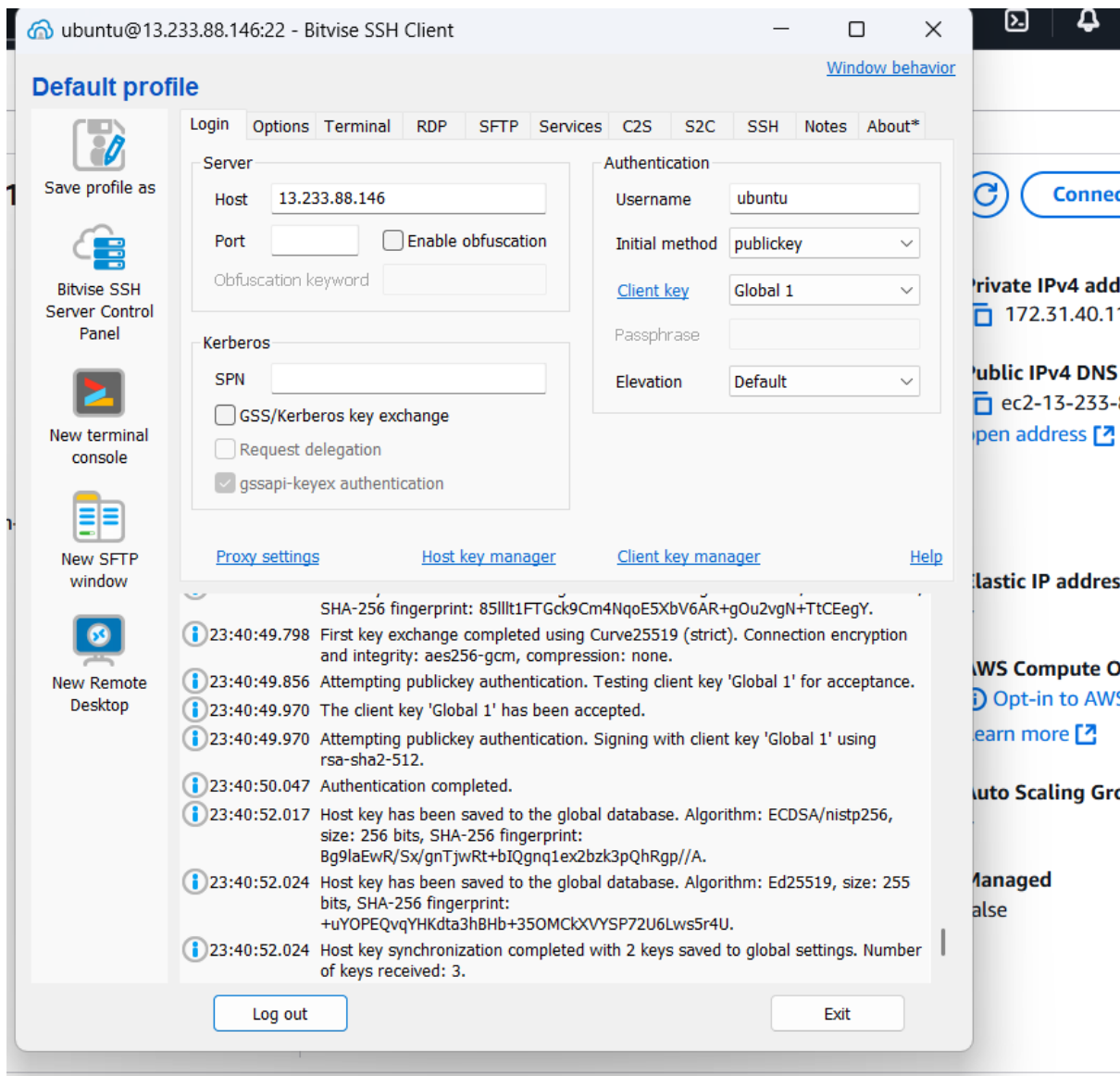
Step 6: Open instances and click of the instance Id of the instance. Copy the IPv4 address.

The screenshot displays the AWS Management Console interface for an EC2 instance. The left sidebar shows the navigation menu with categories like EC2, Instances, Images, Elastic Block Store, and Network & Security. The main content area is titled 'Instance summary for i-082e72c437dba9b4a' and shows various details about the instance, including its ID, IP addresses, hostname, DNS names, and instance type. A tooltip is visible over the 'Public IPv4 address' field, indicating that the address has been copied. The instance is currently in a 'Running' state.

Property	Value
Instance ID	i-082e72c437dba9b4a
IPV6 address	-
Hostname type	IP name: ip-172-31-56-198.ap-south-1.compute.internal
Answer private resource DNS name	IPv4 (A)
Auto-assigned IP address	13.233.198.99 [Public IP]
IAM Role	-
IMDSv2	Required
Operator	-
Instance state	Running
Private IP DNS name (IPv4 only)	ip-172-31-56-198.ap-south-1.compute.internal
Instance type	t2.micro
VPC ID	vpc-094bdb10649d296a9
Subnet ID	subnet-03bae974c39991947
Instance ARN	arn:aws:ec2:ap-south-1:954976318775:instance/i-082e72c437dba9b4a
Private IPv4 addresses	172.31.36.198
Public IPv4 DNS	ec2-13-233-198-99.ap-south-1.compute.amazonaws.com
Elastic IP addresses	-
AWS Compute Optimizer finding	Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto Scaling Group name	-
Managed	false

Step 7:

Open Bitwise SSH Client and paste the IP address in host and set up authentication using the key pair then login and after logging in open new terminal window.

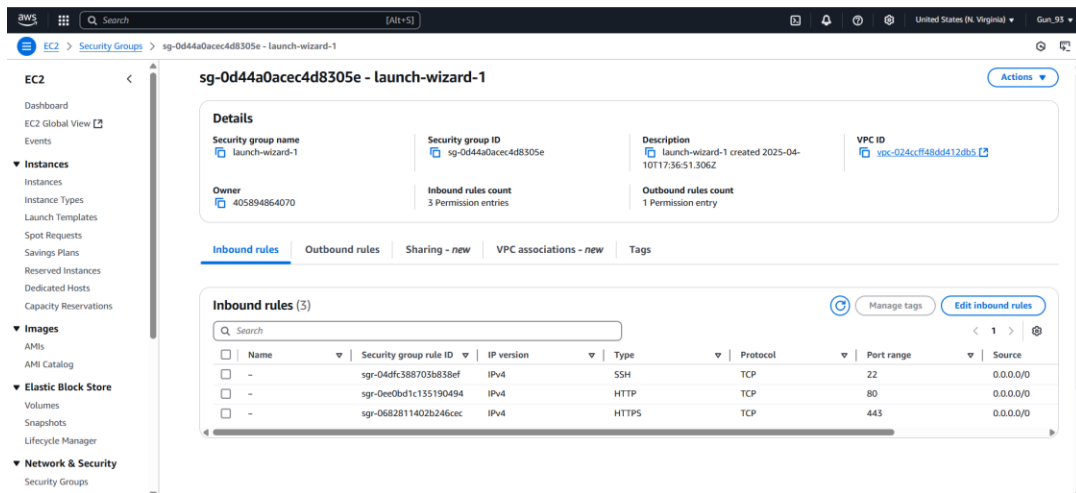


Step 8:

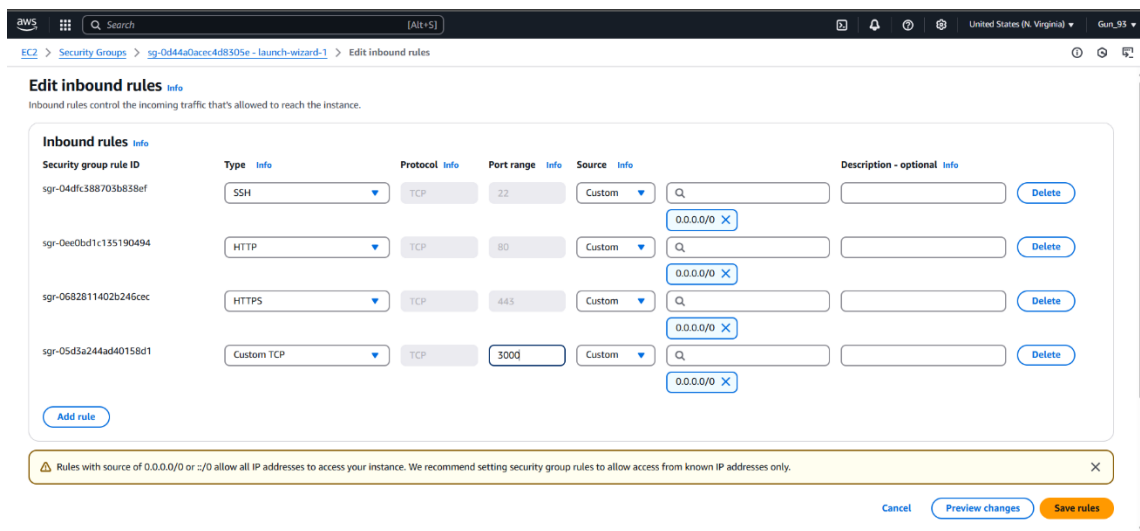
Now clone the GitHub repository from GitHub using command `git clone` paste the link of the repository. Now execute `Sudo apt install node.js` and `npm install`.

```
ubuntu@ip-172-31-24-135:~$ git clone https://github.com/Gun-93/gun93.git
Cloning into 'gun93'...
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 0), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (5/5), 47.89 KiB | 23.94 MiB/s, done.
ubuntu@ip-172-31-24-135:~$
```

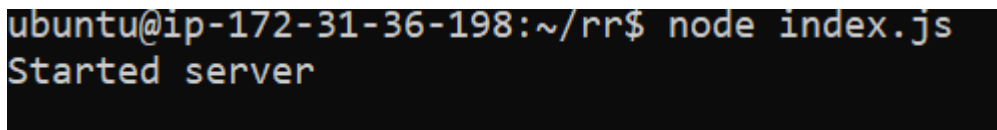
Step 9: Now open AWS and open the current instance then scroll down and go to security Section. Then click on the link in security groups then click on edit inbound rules.



Step 10: Click on add rule and select the type of new rule custom TCP Port range 4000 source 0.0.0.0 or anywhere then click on save rule.



Step 11: Open the Bitwise SSH Client terminal console and start the server using command node index.js



Step 12: Now copy the IPv4 address of the instance and paste it on web browser.

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

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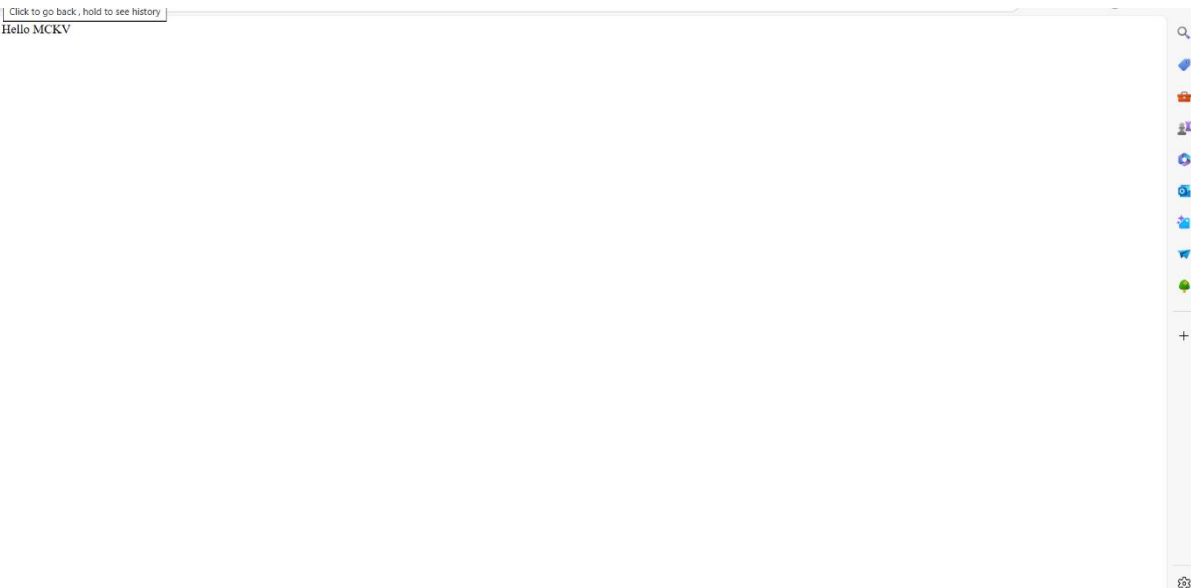
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Step 13:

Then, put the port like this IP address:

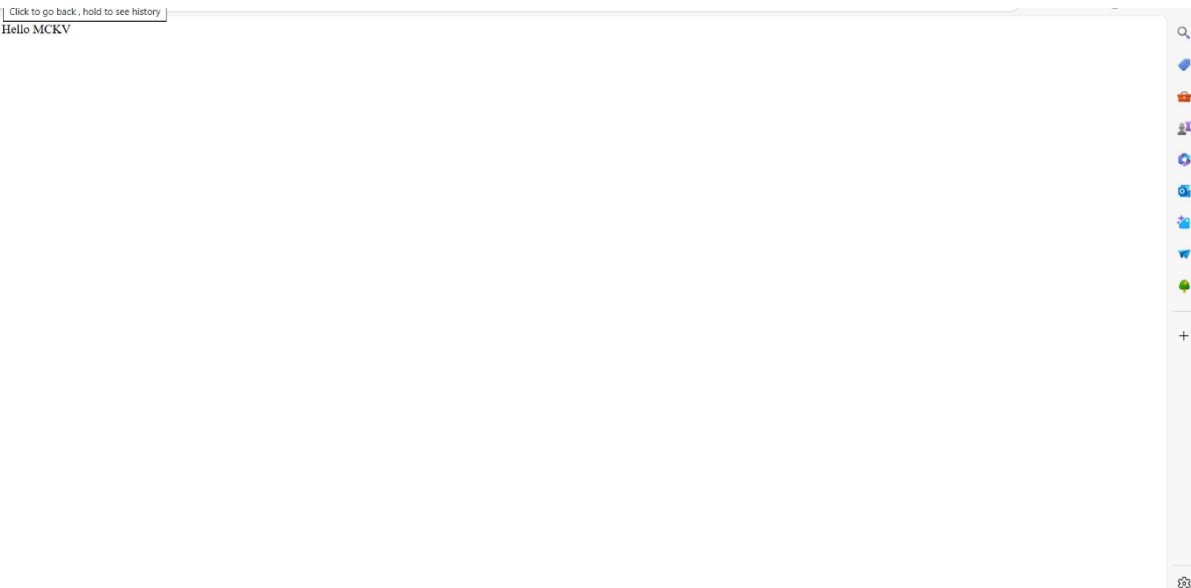
A screenshot of a web browser window. The address bar shows a URL that has been partially redacted with a grey box. Below the address bar, the page content displays the text "Hello MCKV". The browser's sidebar on the right is visible, showing various extension icons and a "+" button at the bottom. The page has a light blue header bar at the top.

Click to go back, hold to see history

Hello MCKV

Step 13:

Then, put the port like this IP address:

A screenshot of a web browser window. The address bar shows a URL that has been partially redacted with a grey box. Below the address bar, the page content displays the text "Hello MCKV". The browser's sidebar on the right is visible, showing various extension icons and a "+" button at the bottom. The page has a light blue header bar at the top.

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